

Glaucoma

Guidelines for Clinical Case Presentations

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HISTORY:

- Diminished Vision
- Pain
 - Time of Day
 - Prone Position
 - Relieved by sleep
 - Emotional Factor
 - Associated Redness
- Watering
- Nausea and Vomiting
- Coloured Haloes
- Early Onset, Frequent change of glasses

Relevant History

- Trauma
- Intraocular Surgery (including previous surgeries for glaucoma)
- Use of topical medication
- Bronchial Asthma: contraindication for Beta-blockers
- Any cause for steroid use – note type and duration
- Thyroid Eye Disease
- Diabetes and Hypertension

Family history of glaucoma

Medical History

Personal History

EXAMINATION:

- General Physical Examination
- Skin: naevi, neurofibromatosis (or any phakomatosis)
- Skeletal: Facial and Dental
- CVS: PDA associated with Rubella
- CNS: phakomatosis
- RS: bronchial asthma, steroid use, beta-blocker contraindication

Ocular Examination

- Head posture
- Ocular Posture
- Visual Acuity
 - Best corrected vision for distance and near
 - Colour
 - Confrontation
- Lids and Adnexa
 - Hyperpigmentation
 - Port-wine stain
- Conjunctiva
 - Follicles
 - Adrenochrome pigmentation
 - Presence of Bleb (cicatrical or filtering)

- Cornea
 - Size, Shape, CCC
 - Corneal Edema
 - Healed opacities: longstanding edema
 - Vascularisation
 - Pigments: Kruckenberg spindle
 - Keratic Precipitates
- Anterior Chamber: Gonioscopic examination presented later
 - Depth (central and peripheral)
 - Irido-corneal touch
 - Cells and Flare
- Iris
 - Pattern, NVI, Atrophy, polycoria, Ectropion uveae, Pseudoexfoliation, Heterochromia, Lisch nodules, Peripheral Iridectomy (Patent vs. Non-patent)
 - Synechiae: peripheral anterior and posterior
- Pupil: Direct and consensual
- Lens
 - Glaucomflecken, pseudoexfoliation, Vossius' ring, lens subluxation
- Retrolental cells
- Fundus examination:
 - Distant direct ophthalmoscopy
 - Direct ophthalmoscopy (media, disc)
 - Disc
 - Size
 - Shape
 - Margins
 - C/D ratio (Horizontal and Vertical)
 - Cup: depth, lamellar dot sign
 - Neuro-retinal rim (colour and thinning, capillary count)
 - Vessles on the disc: tortuosity, nasal shift, bayoneting, bleeds)
 - Peripapillary defects
 - NFL defects
 - Retinal vessels, fundus periphery and macula - look for the possibility of neovascularisation
- Intraocular pressure
- Gonioscopy (draw a diagram if possible)
 - visible angle structures
 - iris configuration (occludable angle)
 - peripheral anterior synichiae
 - neovascularisation
 - Other findings:
 - Pigmentation, Sampolesi's line, Loch Ness monster, Lister's morning mist
- Fields
- Automated perimetry if possible

van Herick's Grading:

1. PAC < ¼ CT
2. PAC = ¼ CT
3. PAC = ½ to ¼ CT
4. PAC > 1 CT (normal depth)

Scheie's:

0. Wide open:
 - Ciliary body visualised
1. Open angle with which at least scleral spur can be visualised
2. Only trabeculum can be identified
3. Only Schwalbe's line can be identified
4. Closed:
 - Irido-corneal contact.

Shaffer's:

- A. Grade 4: (35 to 45 degrees)
- B. Grade 3: (25 to 35 degrees)
- C. Grade 2: (10 to 25 degrees)
- D. Grade 1: (0 to 10 degrees)
- Grade 0: (0 degrees)

Spaeth's:

Based on three factors:

- Angular width of angle recess (10 to 50 degrees)
- Configuration of peripheral iris (S, R, Q)
- Apparent insertion of iris root (A to E)